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Heat treatment vessel coupler, heat treatment vessel coupling method, and heat treatment method

Abstract

A heat treatment vessel coupler for coupling together heat treatment vessels adjacent in a horizontal direction, of a plurality of rectangular box-shaped heat treatment vessels each loaded with a material to be treated, conveyed in a conveying direction on a plurality of rotationally driven conveyor rollers arranged in parallel inside a heat treatment furnace, the coupler comprises a plate body smaller than each of the heat treatment vessels, at least a pair of engagement protrusions each projecting from the plate body and engaging with a corresponding one of side walls of the heat treatment vessels, and a spacer protrusion projecting from the plate body between the at least a pair of engagement protrusions so as to be sandwiched between the heat treatment vessels with which the at least a pair of engagement protrusions engage respectively.

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Background/Summary**TECHNICAL FIELD**

(1) The present invention relates to a heat treatment vessel coupler for coupling together rectangular box-shaped heat treatment vessels conveyed in one direction on a plurality of conveyor rollers arranged within a heat treatment furnace, and relates to a technique restricting deterioration in circulation of atmosphere within the heat treatment furnace while suppressing meandering of the heat treatment vessels and variability in progress of the heat treatment vessels.

BACKGROUND ART

(2) A rectangular box-shaped heat treatment vessel is well known that is conveyed in a conveying direction, with a material to be treated mounted thereon, on a plurality of rotationally driven conveyor rollers arranged in parallel inside a heat treatment furnace. An example thereof is a heat treatment vessel described in Patent Document 1. Though not the heat treatment vessel, a coupler is well known that couples flat plate-shaped heat treatment setters together. An example thereof is a coupler depicted in FIGS. 6 to 10 of Patent Document 2. In Patent Document 2, the front and rear treatment setters are coupled together in contact with each other without any gap therebetween, so that relative displacement between the heat treatment setters is constrained to suppress meandering of the heat treatment setters.

(3) By applying the coupler technique of Patent Document 2 to the heat treatment vessel of Patent Document 1, a heat treatment vessel coupler may be conceivable that couples the front and rear heat treatment vessels in contact with each other without any gap therebetween. However, the heat treatment vessel coupler described above involves problems, due to the gap-free contact between the heat treatment vessels, that the front and rear heat treatment vessels become difficult to be separated from each other after heat treatment and that atmospheric circulation becomes poor between above and below the heat treatment vessels within a heat treatment furnace in operation, resulting in, for example, deterioration of the treating efficiency to treat the material to be treated by heat.

(4) As described in Patent Document 3, a type is known where a plurality of products to be fired are arranged in a direction orthogonal to a conveying direction, the products to be fired being conveyed on a plurality of rotationally driven conveyor rollers arranged in parallel inside a heat treatment furnace. This Patent Document 3 entails a problem, for example, that since the conveyor rollers bend downward at their central portions due to their own weights and loads of the products to be

fired, the products to be fired lying at central portions of the conveyor rollers, of the products to be fired conveyed in the conveying direction, are delayed in progress with respect to the products to be fired lying on both end sides of the conveyor rollers, hindering automatic unloading at the furnace outlet. Though it is also conceivable in Patent Document 3 to use rectangular box-shaped heat treatment vessels instead of the products to be fired, the same problem as the above arises that the heat treatment vessels at central portions are delayed in progress with respect to the heat treatment vessels at both end sides, i.e., a problem arises that the heat treatment vessels progress differently. Difference in progress between the heat treatment vessels leads to, for example, extension of time during which a shutter disposed at the outlet of the heat treatment furnace is opened, causing e.g. another problem of increased inflow of external air into the heat treatment furnace.

PRIOR ART DOCUMENTS

Patent Documents

(5) Patent Document 1: JP2014122783A Patent Document 2: JP2000111268A Patent Document 3: JPS6143198Y2 Patent Document 4: JPS437173Y1

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(6) On the other hand, a heat treatment vessel coupler as described in Patent Document 4 is known. The heat treatment vessel coupler of Patent Document 4 is a lid-like coupler that couples adjacent rectangular box-shaped heat treatment vessels together and closes openings of the adjacent heat treatment vessels. Therefore, if the lid-like coupler is used for the heat treatment vessels adjacent in the direction orthogonal to the conveying direction, the relative displacement between the adjacent heat treatment vessels is constrained by the lid-like coupler so that the variability in progress of the heat treatment vessels can be suppressed. However, the lid-like coupler as described above not only closes the openings of the adjacent heat treatment vessels but also closes the space defined between the adjacent heat treatment vessels, which deteriorate the atmospheric circulation between above and below the heat treatment vessels, resulting in e.g. a problem of deterioration in treatment efficiency to treat the material to be treated by heat.

(7) The present invention was conceived against a background of the above circumstances and its object is to provide a heat treatment vessel coupler capable of easily separating heat treatment vessels after heat treatment and suppressing deterioration in atmospheric circulation within a heat treatment furnace, while restricting meandering of the heat treatment vessels or variability in progress between the heat treatment vessels.

(8) According to a first aspect of the invention, (a) there is provided a heat treatment vessel coupler for coupling together heat treatment vessels adjacent in a horizontal direction, of a plurality of rectangular box-shaped heat treatment vessels each loaded with a material to be treated, conveyed in a conveying direction on a plurality of rotationally driven conveyor rollers arranged in parallel inside a heat treatment furnace, the coupler comprises (b) a plate body smaller than each of the heat treatment vessels, (c) at least a pair of engagement protrusions each projecting from the plate body and engaging with a corresponding one of side walls of the heat treatment vessels, and (d) a spacer protrusion projecting from the plate body between the at least a pair of engagement protrusions so as to be sandwiched between the heat treatment vessels with which the at least a pair of engagement protrusions engage respectively.

(9) According to a second aspect of the invention, in the heat treatment vessel coupler according to the first aspect of the invention, (a) the coupler couples mutually the heat treatment vessels adjacent in the conveying direction, of the plurality of rectangular box-shaped heat treatment vessels or couples mutually the heat treatment vessels adjacent in the conveying direction and mutually the heat treatment vessels adjacent in a direction orthogonal to the conveying direction, (b) the at least a pair of engagement protrusions each engage with a corresponding one of corners defined between side walls of the heat treatment vessels, and (c) the spacer protrusion projects from the plate body

between the at least a pair of engagement protrusions so as to be sandwiched between the corners defined between side walls of the heat treatment vessels with which the at least a pair of engagement protrusions engage respectively.

(10) According to a third aspect of the invention, in the heat treatment vessel coupler according to the second aspect of the invention, (a) the at least a pair of engagement protrusions include a total of four engagement protrusions, i.e., a pair of engagement protrusions each engaging with a corresponding one of corners defined between side walls of a pair of heat treatment vessels adjacent in the conveying direction and a pair of engagement protrusions each engaging with a corresponding one of corners defined between side walls of a pair of heat treatment vessels adjacent in a direction orthogonal to the conveying direction, and (b) the spacer protrusion projects from the plate body between the four engagement protrusions so as to be sandwiched between the pair of heat treatment vessels adjacent in the conveying direction and between the pair of heat treatment vessels adjacent in the direction orthogonal to the conveying direction.

(11) According to a fourth aspect of the invention, in the heat treatment vessel coupler according to the third aspect of the invention, the spacer protrusion comprises a conveying-direction gap forming portion intervening between heat treatment vessels adjacent in the conveying direction to form a gap therebetween; and an orthogonal-direction gap forming portion intervening between heat treatment vessels adjacent in a direction orthogonal to the conveying direction to form a gap therebetween.

(12) According to a fifth aspect of the invention, in the heat treatment vessel coupler according to the fourth aspect of the invention, (a) the conveying-direction gap forming portion is elongated in a direction orthogonal to the conveying direction, (b) the orthogonal-direction gap forming portion is elongated in the conveying direction, and (c) the spacer protrusion is a protrusion including the conveying-direction gap forming portion and the orthogonal-direction gap forming portion that intersect each other at right angles.

(13) According to a sixth aspect of the invention, in the heat treatment vessel coupler according to the second aspect of the invention, (a) the at least a pair of engagement protrusions include two engagement protrusions each engaging with a corresponding one of corners defined between side walls of a pair of heat treatment vessels adjacent in the conveying direction, (b) the spacer protrusion projects from the plate body between the two engagement protrusions so as to be sandwiched between the pair of heat treatment vessels adjacent in the conveying direction.

(14) According to a seventh aspect of the invention, in the heat treatment vessel coupler according to the first aspect of the invention, (a) the coupler couples mutually heat treatment vessels adjacent in a direction orthogonal to the conveying direction, of the rectangular box-shaped heat treatment vessels, (b) the pair of engagement protrusions project integrally from the plate body and engage respectively with one side wall, lying toward a heat treatment vessel on the other hand, of side walls of a heat treatment vessel on one hand, of the heat treatment vessels adjacent in the orthogonal direction and with the other side wall, facing the one side wall, of side walls of a heat treatment vessel on the other hand, of the heat treatment vessels adjacent in the orthogonal direction, and (c) the spacer protrusion project from the plate body between the pair of engagement protrusions so as to be sandwiched between the one side wall and the other side wall.

(15) According to an eighth aspect of the invention, in the heat treatment vessel coupler according to the seventh aspect of the invention, the pair of engagement protrusions and the spacer protrusion are elongated in the conveying direction and are parallel to one another.

(16) According to a ninth aspect of the invention, in the heat treatment vessel coupler according to the seventh or eighth aspects of the invention, the pair of engagement protrusions engage with an end of the one side wall and an end of the other side wall, respectively.

(17) According to a tenth aspect of the invention, in the heat treatment vessel coupler according to the seventh or eighth aspects of the invention, in a state where the coupler couples adjacent vessels together, the plate body is an elongated plate extending along the one side wall and the other side

wall, one of the pair of engagement protrusions being a longitudinal protrusion projecting from a side edge, lying toward the heat treatment vessel on one hand, of the elongated plate, so as to face the one side wall, the other of the pair of engagement protrusions being a longitudinal protrusion projecting from a side edge, lying toward the heat treatment vessel on the other hand, of the elongated plate, so as to face the other side wall.

(18) According to a eleventh aspect of the invention, in the heat treatment vessel coupler according to the ninth aspect of the invention, a misalignment prevention protrusion projecting on the plate body from a side edge not having the pair of engagement protrusions, of both side edges of the plate body, to a direction identical to a direction of projection of the pair of engagement protrusions, the misalignment prevention protrusion being positioned, with intervals greater than a thickness of the side walls of the heat treatment vessel, between the pair of engagement protrusions.

(19) According to a twelfth aspect of the invention, (a) a heat treatment vessel coupling method of coupling mutually, by use of a coupler, heat treatment vessels adjacent in a conveying direction or heat treatment vessels adjacent in an orthogonal direction orthogonal to the conveying direction, of rectangular box-shaped heat treatment vessels each loaded with a material to be treated, conveyed on a plurality of rotationally driven conveyor rollers arranged in parallel inside a heat treatment furnace, (b) the coupler comprises (b1) a plate body smaller than each of the heat treatment vessels, (b2) a pair of engagement protrusions projecting integrally from the plate body and engaging respectively with one side wall, lying toward a heat treatment vessel on the other hand, of side walls of a heat treatment vessel on one hand, of heat treatment vessels adjacent in the conveying direction or the orthogonal direction and with the other side wall, facing the one side wall, of side walls of the heat treatment vessel on the other hand, of the heat treatment vessels adjacent in the conveying direction or the orthogonal direction, and (b3) a spacer protrusion projecting from the plate body between the pair of engagement protrusions so as to be sandwiched between the one side wall and the other side wall.

(20) According to a thirteenth aspect of the invention, (a) a heat treatment method using a heat treatment vessel coupler, the coupler coupling mutually, heat treatment vessels adjacent in a conveying direction of the heat treatment vessels or heat treatment vessels adjacent in an orthogonal direction orthogonal to the conveying direction, of rectangular box-shaped heat treatment vessels each loaded with a material to be treated, conveyed on a plurality of rotationally driven conveyor rollers arranged in parallel inside a heat treatment furnace, (b) the coupler comprises (b1) a plate body smaller than each of the heat treatment vessels; (b2) a pair of engagement protrusions projecting integrally from the plate body and engaging respectively with one side wall, lying toward a heat treatment vessel on the other hand, of side walls of a heat treatment vessel on one hand, of heat treatment vessels adjacent in the conveying direction or the orthogonal direction and with the other side wall, facing the one side wall, of side walls of the heat treatment vessel on the other hand, of the heat treatment vessels adjacent in the conveying direction or the orthogonal direction, and (b3) a spacer protrusion projecting from the plate body between the pair of engagement protrusions so as to be sandwiched between the one side wall and the other side wall.

(21) According to the first aspect of the invention, the coupler includes (b) a plate body smaller than each of the heat treatment vessels, (c) a pair of engagement protrusions projecting integrally from the plate body and engaging respectively with one side wall, and (d) a spacer protrusion projecting from the plate body between the pair of engagement protrusions so as to be sandwiched between the one side wall and the other side wall. Consequently, during the conveyance, engagement of the pair of engagement protrusions of the coupler restricts mutual displacement of the heat treatment vessels between the heat treatment vessels adjacent in the conveying direction or between the heat treatment vessels adjacent in the direction orthogonal to the conveying direction, thereby enabling suppression of meandering of the heat treatment vessels or suppression of

variability in progress between the heat treatment vessels. By sandwiching the spacer protrusion between the heat treatment vessels, the close proximity is restricted between the heat treatment vessels so that the heat treatment vessels are conveyed inside the heat treatment furnace, with the gap formed between the heat treatment vessels, thereby facilitating separation of the heat treatment vessels from each other after heat treatment and achieving suppression of deterioration in the atmospheric circulation within the heat treatment furnace.

(22) According to the second aspect of the invention, (a) the coupler couples mutually the heat treatment vessels adjacent in the conveying direction, of the plurality of rectangular box-shaped heat treatment vessels or couples mutually the heat treatment vessels adjacent in the conveying direction and mutually the heat treatment vessels adjacent in a direction orthogonal to the conveying direction, (b) the at least a pair of engagement protrusions each engage with a corresponding one of corners defined between side walls of the heat treatment vessels, and (c) the spacer protrusion projects from the plate body between the at least a pair of engagement protrusions so as to be sandwiched between the corners defined between side walls of the heat treatment vessels with which the at least a pair of engagement protrusions engage respectively. As a result, the same effect as in the first invention can be obtained in coupling mutually the heat treatment vessels adjacent in the conveying direction or in coupling mutually the heat treatment vessels adjacent in the conveying direction and mutually the heat treatment vessels adjacent in the direction orthogonal to the conveying direction. The at least a pair of engagement protrusions engage with corners of side walls of the heat treatment vessels. Since the corners of side walls of the heat treatment vessels have a high strength, the durability of the heat treatment vessels increases.

(23) According to the third aspect of the invention, (a) the at least a pair of engagement protrusions include a total of four engagement protrusions, i.e., a pair of engagement protrusions each engaging with a corresponding one of corners defined between side walls of a pair of heat treatment vessels adjacent in the conveying direction and a pair of engagement protrusions each engaging with a corresponding one of corners defined between side walls of a pair of heat treatment vessels adjacent in a direction orthogonal to the conveying direction, and (b) the spacer protrusion projects from the plate body between the four engagement protrusions so as to be sandwiched between the pair of heat treatment vessels adjacent in the conveying direction and between the pair of heat treatment vessels adjacent in the direction orthogonal to the conveying direction. Consequently, mutual displacement of the heat treatment vessels between the heat treatment vessels adjacent in the conveying direction or between the heat treatment vessels adjacent in the direction orthogonal to the conveying direction, thereby enabling suppression of meandering of the heat treatment vessels by the single coupler.

(24) According to the fourth aspect of the invention, the spacer protrusion comprises a conveying-direction gap forming portion intervening between heat treatment vessels adjacent in the conveying direction to form a gap therebetween; and an orthogonal-direction gap forming portion intervening between heat treatment vessels adjacent in a direction orthogonal to the conveying direction to form a gap therebetween. Consequently, the close proximity is restricted between the heat treatment vessels so that the heat treatment vessels are conveyed inside the heat treatment furnace, with the gap formed between the heat treatment vessels.

(25) According to the fifth aspect of the invention, (a) the conveying-direction gap forming portion is elongated in a direction orthogonal to the conveying direction, (b) the orthogonal-direction gap forming portion is elongated in the conveying direction, and (c) the spacer protrusion is a protrusion including the conveying-direction gap forming portion and the orthogonal-direction gap forming portion that intersect each other at right angles. Consequently, the close proximity is restricted between the heat treatment vessels, so that the heat treatment vessels are conveyed inside the heat treatment furnace, with the gap formed between the heat treatment vessels.

(26) According to a sixth aspect of the invention, (a) the at least a pair of engagement protrusions

include two engagement protrusions each engaging with a corresponding one of corners defined between side walls of a pair of heat treatment vessels adjacent in the conveying direction, (b) the spacer protrusion projects from the plate body between the two engagement protrusions so as to be sandwiched between the pair of heat treatment vessels adjacent in the conveying direction. Consequently, mutual displacement of the heat treatment vessels between the heat treatment vessels adjacent in the conveying direction the close proximity is restricted between the heat treatment vessel, so that the heat treatment vessels are conveyed inside the heat treatment furnace, with the gap formed between the heat treatment vessels.

(27) According to the seventh aspect of the invention, (a) the coupler couples mutually heat treatment vessels adjacent in a direction orthogonal to the conveying direction, of the rectangular box-shaped heat treatment vessels, (b) the pair of engagement protrusions project integrally from the plate body and engage respectively with one side wall, lying toward a heat treatment vessel on the other hand, of side walls of a heat treatment vessel on one hand, of the heat treatment vessels adjacent in the orthogonal direction and with the other side wall, facing the one side wall, of side walls of a heat treatment vessel on the other hand, of the heat treatment vessels adjacent in the orthogonal direction, and (c) the spacer protrusion project from the plate body between the pair of engagement protrusions so as to be sandwiched between the one side wall and the other side wall. Hence, if the heat treatment vessels are each about to progress differently, the one side wall of the heat treatment vessel on one hand or the other side wall of the heat treatment vessel on the other engage with a corresponding one of the pair of engagement protrusions so that the mutual displacement is restricted, thus enabling suppression of variability in progress between the heat treatment vessels. By sandwiching the spacer protrusion between the one side wall of the heat treatment vessel on one hand and the other side wall of the heat treatment vessel on the other, the close proximity is restricted between the heat treatment vessel on one hand and the heat treatment vessel on the other, so that the heat treatment vessels are conveyed inside the heat treatment furnace, with the gap formed between the heat treatment vessel on one hand and the heat treatment vessel on the other that are adjacent to each other, thereby achieving suppression of deterioration in the atmospheric circulation within the heat treatment furnace.

(28) According to the eighth aspect of the invention, the pair of engagement protrusions and the spacer protrusion are elongated in the conveying direction and are parallel to one another. Consequently, by the elongate spacer protrusion, the close proximity is restricted between the heat treatment vessels.

(29) According to the ninth aspect of the invention, the pair of engagement protrusions engage with an end of the one side wall and an end of the other side wall, respectively. Consequently, by the elongate spacer protrusion, the close proximity is restricted between the heat treatment vessels. For this reason, if the heat treatment vessels are each about to progress differently, the corner of the heat treatment vessel comes into abutment against a corresponding one of the pair of engagement protrusions, so that mutual displacement can properly be restricted between the heat treatment vessels adjacent in the orthogonal direction.

(30) According to the tenth aspect of the invention, (a) in a state where the coupler couples adjacent vessels together, (b) the plate body is an elongated plate extending along the one side wall and the other side wall, (c) one of the pair of engagement protrusions being a longitudinal protrusion projecting from a side edge, lying toward the heat treatment vessel on one hand, of the elongated plate, so as to face the one side wall, (d) the other of the pair of engagement protrusions being a longitudinal protrusion projecting from a side edge, lying toward the heat treatment vessel on the other hand, of the elongated plate, so as to face the other side wall. For this reason, if the heat treatment vessels adjacent in the orthogonal direction are mutually displaced, the corner of the heat treatment vessel comes into abutment against a corresponding one of the pair of engagement protrusions, so that mutual displacement can properly be restricted between the heat treatment vessels adjacent in the orthogonal direction.

(31) According to the eleventh aspect of the invention, the coupler comprises a misalignment prevention protrusion projecting on the plate body from a side edge not having the pair of engagement protrusions, of both side edges of the plate body, to a direction identical to a direction of projection of the pair of engagement protrusions. The misalignment prevention protrusion is positioned, with intervals greater than a thickness of the side walls of the heat treatment vessel, between the pair of engagement protrusions. Hence, the misalignment prevention protrusion can prevent misalignment of the couplers by sandwiching the side walls of the heat treatment vessels between the misalignment prevention protrusion and the pair of engagement protrusions.

(32) According to the twelfth aspect of the invention, heat treatment vessels adjacent in a conveying direction or heat treatment vessels adjacent in an orthogonal direction orthogonal to the conveying direction are coupled together by use of a coupler including (a) a plate body smaller than each of the heat treatment vessels, (b) a pair of engagement protrusions projecting integrally from the plate body and engaging respectively with one side wall, lying toward a heat treatment vessel on the other hand, of side walls of a heat treatment vessel on one hand, of heat treatment vessels adjacent in the conveying direction or the orthogonal direction and with the other side wall, facing the one side wall, of side walls of the heat treatment vessel on the other hand, of the heat treatment vessels adjacent in the conveying direction or the orthogonal direction, and (c) a spacer protrusion projecting from the plate body between the pair of engagement protrusions so as to be sandwiched between the one side wall and the other side wall. Consequently, during the conveyance, engagement of the pair of engagement protrusions of the coupler restricts mutual displacement of the heat treatment vessels between the heat treatment vessels adjacent in the conveying direction or between the heat treatment vessels adjacent in the direction orthogonal to the conveying direction, thereby enabling suppression of meandering of the heat treatment vessels or suppression of variability in progress between the heat treatment vessels. By sandwiching the spacer protrusion between the heat treatment vessels, the close proximity is restricted between the heat treatment vessels, so that the heat treatment vessels are conveyed inside the heat treatment furnace, with the gap formed between the heat treatment vessels, thereby facilitating separation of the heat treatment vessels from each other after heat treatment and achieving suppression of deterioration in the atmospheric circulation within the heat treatment furnace.

(33) According to a heat treatment method of the thirteenth aspect of the invention, heat treatment is performed in the state where the heat treatment vessels adjacent in the conveying direction of the heat treatment vessels or the heat treatment vessels adjacent in the orthogonal direction orthogonal to the conveying direction are coupled together by a heat treatment vessel coupler including (a) a plate body smaller than each of the heat treatment vessels, (b) a pair of engagement protrusions projecting integrally from the plate body and engaging respectively with one side wall, lying toward a heat treatment vessel on the other hand, of side walls of a heat treatment vessel on one hand, of heat treatment vessels adjacent in the conveying direction or the orthogonal direction and with the other side wall, facing the one side wall, of side walls of the heat treatment vessel on the other hand, of the heat treatment vessels adjacent in the conveying direction or the orthogonal direction, and (c) a spacer protrusion projecting from the plate body between the pair of engagement protrusions so as to be sandwiched between the one side wall and the other side wall. In consequence, during the conveyance of the heat treatment vessels, engagement of the pair of engagement protrusions of the coupler restricts mutual displacement of the heat treatment vessels between the heat treatment vessels adjacent in the conveying direction or between the heat treatment vessels adjacent in the direction orthogonal to the conveying direction, thereby enabling suppression of meandering of the heat treatment vessels or suppression of variability in progress between the heat treatment vessels. By sandwiching the spacer protrusion between the heat treatment vessels, the close proximity is restricted between the heat treatment vessels, so that the heat treatment vessels are conveyed inside the heat treatment furnace, with the gap formed between the heat treatment vessels, thereby facilitating separation of the heat treatment vessels from each

other after heat treatment and achieving suppression of deterioration in the atmospheric circulation within the heat treatment furnace.

Description

- (1) FIG. 1 is a perspective view explaining a schematic configuration of a continuous conveyor type heat treatment furnace to which the present invention is properly applied.
- (2) FIG. 2 is a view showing the state where the heat treatment vessels coupled together by the coupler as shown in FIG. 1 are stacked in e.g. two layers.
- (3) FIG. 3 is an enlarged view of the coupler shown in FIG. 1.
- (4) FIG. 4 is a view of the coupler 18 of FIG. 3, seen from the direction of arrow A.
- (5) FIG. 5 is a perspective view explaining a schematic configuration of a heat treatment furnace of another embodiment of the present invention.
- (6) FIG. 6 is an enlarged view of one of the pair of first coupler and the second coupler shown in FIG. 5.
- (7) FIG. 7 is a view of the coupler of FIG. 6, seen from the direction of arrow B.
- (8) FIG. 8 is a perspective view explaining a configuration of a coupler disposed in a heat treatment furnace of another embodiment of the present invention.
- (9) FIG. 9 is a view of the coupler of FIG. 8, seen from the direction of arrow C.
- (10) FIG. 10 is a perspective view explaining a configuration of a heat treatment vessel disposed in a heat treatment furnace of another embodiment of the present invention.
- (11) FIG. 11 is a perspective view explaining a schematic configuration of the continuous conveyor type heat treatment furnace to which a coupler of another example (Example 5) of the present invention is properly applied.
- (12) FIG. 12 is an enlarged view of the coupler coupling together the heat treatment vessels adjacent in the orthogonal direction in FIG. 11.
- (13) FIG. 13 is a view of the coupler of FIG. 12, seen from the direction of arrow A.
- (14) FIG. 14 is a perspective view explaining another example (Example 6) of application of the coupler of the present invention.
- (15) FIG. 15 is a perspective view explaining another example (Example 7) of application of the coupler of the present invention.
- (16) FIG. 16 is a perspective view explaining another example (Example 8) of application of the coupler of the present invention.
- (17) FIG. 17 is an enlarged view of one of the pair of couplers coupling together the heat treatment vessel and the heat treatment vessel adjacent in the orthogonal direction.
- (18) FIG. 18 is a view of the coupler of FIG. 17, seen from the direction of arrow B.
- (19) FIG. 19 is a perspective view explaining an example of application of a coupler of another embodiment (Example 9) of the present invention.
- (20) FIG. 20 is an enlarged view of the coupler coupling together the heat treatment vessel and the heat treatment vessel adjacent in the orthogonal direction in FIG. 19.
- (21) FIG. 21 is a view of the coupler of FIG. 20, seen from the direction of arrow C.

MODES FOR CARRYING OUT THE INVENTION

(22) Embodiments of the present invention will hereinafter be described in detail with reference to the drawings. In the following embodiments, the figures are simplified or modified appropriately, and the dimension ratios, shapes, etc. of parts are not necessarily correctly drawn.

First Embodiment

(23) FIG. 1 is a perspective view explaining a schematic configuration of a continuous conveyor type heat treatment furnace 12 to which the present invention is properly applied. The heat treatment furnace 12 is a roller hearth kiln that includes a plurality of conveyor rollers 16 to convey

heat treatment vessels **14** continuously and rectilinearly and that performs heat treatment of a powder-like or pellet-like material to be treated accommodated or placed within the heat treatment vessels **14** at a specified temperature under a predetermined in-furnace atmosphere, e.g., an inert gas atmosphere. The conveyor rollers **16** are each a sintered body of a heat-resistant inorganic material such as Al.sub.2O.sub.3 or SiC. The heat treatment furnace **12** includes, as shown in FIG. **1**: the plurality of conveyor rollers **16** each having a cylindrical shape with the same diameter, horizontally arranged in parallel and at equal intervals inside the heat treatment furnace **12**; and a driving device (not shown) rotationally driving the conveyor rollers **16** in synchronism with one another. The heat treatment furnace **12** is configured such that the heat treatment vessels **14** arranged on the conveyor rollers **16** are conveyed in a horizontal direction i.e. a conveying direction **F1**. The material to be treated **10** is for example a battery material, and is preferably an all-solid-state battery electrode material or a lithium-ion battery electrode material.

(24) For example, inside the heat treatment furnace **1**, as shown in FIG. **1**, the heat treatment vessels **14** are conveyed by fours in the conveying direction **F1**, each four heat treatment vessels **14** being coupled together in the conveying direction **F1** and an orthogonal direction **F2** by a coupler **18**. The four heat treatment vessels **14** shown in FIG. **1** are each of the same shape i.e. a rectangular box shape and are each formed of a sintered body obtained by: integrally molding a heat-resistant inorganic material by e.g. pressing; and then sintering the mold. Examples of the heat-resistant inorganic material include mullite-cordierite, mullite, cordierite, alumina, spinel, SiC, and aluminum titanate. Since the coupler **18** is also formed of e.g. the sintered body of the heat-resistant inorganic material, a large frictional resistance occurs between the coupler **18** and the heat treatment vessels **14**. The orthogonal direction **F2** is the longitudinal direction of the conveyor rollers **16** and is parallel to axes of the conveyor rollers **16**, with the conveying direction **F1** being orthogonal to the axes of the conveyor rollers **16**.

(25) The four heat treatment vessels **14** each integrally include, as shown in FIG. **1**: a square bottom wall **14a**; a front wall (side wall) **14b** vertically extending from the bottom wall **14a** at its front end in the conveying direction **F1**; a rear wall (side wall) **14c** vertically extending from the bottom wall **14a** at its rear end in the conveying direction **F1**; a right wall (side wall) **14d** vertically extending from the bottom wall **14a** at its right end in the conveying direction **F1**; and a left wall (side wall) **14e** vertically extending from the bottom wall **14a** at its left end in the conveying direction **F1**. As shown in FIG. **1**, the front wall **14b**, the rear wall **14c**, the right wall **14d**, and the left wall **14e** each having the same thickness each include a notch **14f** through which gas generated from the material to be treated **10** as a result of heat treatment is discharged from the interior of the heat treatment vessel **14** in the case where the heat treatment vessel **14** is multi-tiered (two-tiered in FIG. **2**) as shown in FIG. **2**. The front wall **14b**, the rear wall **14c**, the right wall **14d**, and the left wall **14e** have their respective top faces **14g** that are formed in the same plane. The top faces **14g** of the front wall **14b**, rear wall **14c**, right wall **14d**, and left wall **14e** are each parallel to the bottom wall **14a**. The heat treatment vessel **14** has four corners **14h** that are formed by the front wall **14b**, rear wall **14c**, right wall **14d**, and left wall **14e**, the four corners **14h** having their respective notches **14i** formed such that a top face **18a** of the coupler **18** is level with the top faces **14g** of the front wall **14b**, rear wall **14c**, right wall **14d**, and left wall **14e** as shown in FIG. **2**, or such that the top face **18a** of the coupler **18** lies closer to the bottom wall **14a** than the top faces **14g** of the front wall **14b**, rear wall **14c**, right wall **14d**, and left wall **14e** lie, in the case where the coupler **18** is assembled to the corners **14h** of the heat treatment vessels **14**. This enables the heat treatment vessels **14** coupled by the coupler **18** to be properly multi-tiered. FIG. **2** is a view showing the state where the heat treatment vessels **14** coupled together by the coupler **18** as shown in FIG. **1** are stacked in e.g. two layers.

(26) The configuration of the coupler **18** will now be described with the four heat treatment vessels **14** of FIG. **1** identified as heat treatment vessels **14A** to **14D**. The heat treatment vessel **14A** is the heat treatment vessel **14** arranged on the right side in the conveying direction **F1**, of the pair of heat

treatment vessels **14** lying opposite to the conveying direction F, of the four heat treatment vessels **14**. The heat treatment vessel **14B** is the heat treatment vessel **14** adjacent to the heat treatment vessel **14A** in the conveying direction F1. The heat treatment vessel **14C** is the heat treatment vessel **14** adjacent to the heat treatment vessel **14B** in the orthogonal direction F2 perpendicular to the conveying direction F1, i.e., the heat treatment vessel **14** arranged on the left side of the heat treatment vessel **14B** in the conveying direction F1. The heat treatment vessel **14D** is the heat treatment vessel **14** adjacent to the heat treatment vessel **14A** in the orthogonal direction F2 perpendicular to the conveying direction F1, i.e., the heat treatment vessel **14** arranged on the left side of the heat treatment vessel **14A** in the conveying direction F1.

(27) The coupler **18** includes integrally, as shown in FIGS. **1**, **3**, and **4**, a plate body **18b**, a pair of first engagement protrusions (engagement protrusions) **18c** and **18d**, a pair of second engagement protrusions (engagement protrusions) **18e** and **18f**, and a spacer protrusion **18g**. The plate body **18b** is a plate member of e.g. a square shape, smaller in size than the bottom walls **14a** of the heat treatment vessels **14A** to **14D**. The pair of first engagement protrusions **18c** and **18d** protrude prismatically from the plate body **18b** at its two corners **18h** so as to engage with the inner side of a corresponding one of the four corners **14h** of the heat treatment vessel **14A** and with the inner side of a corresponding one of the four corners **14h** of the heat treatment vessel **14B**. More specifically, the first engagement protrusion **18c** engages with the inner side of the corner **14h** between the front wall **14b** and the left wall **14e** of the heat treatment vessel **14A**, while the first engagement protrusion **18d** engages with the inner side of the corner **14h** between the rear wall **14c** and the left wall **14e** of the heat treatment vessel **14B**. The pair of second engagement protrusions **18e** and **18f** protrude prismatically from the plate body **18b** at its remaining two corners **18h** so as to engage with a corresponding one of the four corners **14h** of the heat treatment vessel **14C** and with a corresponding one of the four corners **14h** of the heat treatment vessel **14D**. More specifically, the second engagement protrusion **18e** engages with the inner side of the corner **14h** between the rear wall **14c** and the right wall **14d** of the heat treatment vessel **14C**, while the second engagement protrusion **18f** engages with the inner side of the corner **14h** between the front wall **14b** and the right wall **14d** of the heat treatment vessel **14D**. That is, the coupler **18** includes a total of four engagement protrusions, i.e., the pair of first engagement protrusions **18c** and **18d** and the pair of second engagement protrusions **18e** and **18f**, disposed protruding from the plate body **18b**, the pair of first engagement protrusions **18c** and **18d** engaging with the inner side of the corners **14h** of the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1, the pair of second engagement protrusions **18e** and **18f** engaging with the inner side of the corners **14h** of the pair of heat treatment vessels **14C** and **14D** adjacent to the pair of heat treatment vessels **14B** and **14A**, respectively, in the orthogonal direction F2 perpendicular to the conveying direction F1. FIG. **3** is an enlarged view of the coupler **18** shown in FIG. **1**, and FIG. **4** is a view of the coupler **18** of FIG. **3**, seen from the direction of arrow A.

(28) The spacer protrusion **18g** includes, as shown in FIG. **4**, a conveying-direction gap forming portion **18i** and an orthogonal-direction gap forming portion **18j**. The conveying direction gap forming portion **18i** extends between the first engagement protrusions **18c** and **18d** and the second engagement protrusions **18e** and **18f** in the conveying direction F1, to leave on its both sides a space greater than the wall thickness of the side walls **14b** to **14e**. The orthogonal-direction gap forming portion **18j** extends between the first engagement protrusions **18c** and **18d** and the second engagement protrusions **18e** and **18f** in the orthogonal direction F2 to leave on its both sides a space greater than the wall thickness of the side walls **14b** to **14e**. As shown in FIG. **4**, the conveying-direction gap forming portion **18i** has an elongated shape extending prismatically in the orthogonal direction F2 perpendicular to the conveying direction F1, while the orthogonal-direction gap forming portion **18j** has an elongated shape extending prismatically in the conveying direction F1. That is, the spacer protrusion **18g** is a raised portion composed of the conveying-direction gap forming portion **18i** and the orthogonal-direction gap forming portion **18j** that intersect at right

angles to form a plus (+) shape.

(29) As shown in FIGS. 1, 3, and 4, the conveying-direction gap forming portion **18i** protrudes from the plate body **18b** between the pair of first engagement protrusions **18c** and **18d** and between the pair of second engagement protrusions **18e** and **18f**, so as to be sandwiched between the corners **14h** of the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1 and between the corners **14h** of the pair of heat treatment vessels **14D** and **14C** adjacent to the pair of heat treatment vessels **14A** and **14B**, respectively, in the orthogonal direction F2. In other words, the conveying-direction gap forming portion **18i** protrudes from the plate body **18b** between the pair of first engagement protrusions **18c** and **18d** and between the pair of second engagement protrusions **18e** and **18f**, so as to be sandwiched between the corners **14h** of the pair of heat treatment vessels **14A** and **14B** with which the pair of first engagement protrusions **18c** and **18d** engage respectively and between the corners **14h** of the pair of heat treatment vessels **14C** and **14D** with which the pair of second engagement protrusions **18e** and **18f** engage respectively. For this reason, when the coupler **18** is assembled to the corners **14h** of the heat treatment vessels **14A** to **14D** as shown in FIG. 1, the conveying-direction gap forming portion **18i** of the spacer protrusion **18g** intervenes between the corners **14h** of the heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1 and between the corners **14h** of the heat treatment vessels **14C** and **14D** adjacent in the conveying direction F1, to form a gap i.e. a conveying-direction gap S1 between the front wall **14b** of the heat treatment vessel **14A** and the rear wall **14c** of the heat treatment vessel **14B** and between the rear wall **14c** of the heat treatment vessel **14C** and the front wall **14b** of the heat treatment vessel **14D**.

(30) As shown in FIGS. 1, 3, and 4, the orthogonal-direction gap forming portion **18j** protrudes from the plate body **18b** between the first engagement protrusion **18c** and the second engagement protrusion **18f** and between the first engagement protrusion **18d** and the second engagement protrusion **18e**, so as to be sandwiched between the corners **14h** of the pair of heat treatment vessels **14A** and **14D** and between the corners **14h** of the pair of heat treatment vessels **14B** and **14C**. In other words, the orthogonal-direction gap forming portion **18j** protrudes from the plate body **18b** between the first engagement protrusion **18c** and the second engagement protrusion **18f** and between the first engagement protrusion **18d** and the second engagement protrusion **18e**, so as to be sandwiched between the corners **14h** of the pair of heat treatment vessels **14A** and **14D** with which the pair of first engagement protrusions **18c** and **18f** engage respectively and between the corners **14h** of the pair of heat treatment vessels **14B** and **14C** with which the first engagement protrusion **18d** and the second engagement protrusion **18e** engage respectively. For this reason, when the coupler **18** is assembled to the corners **14h** of the heat treatment vessels **14A** to **14D** as shown in FIG. 1, the orthogonal-direction gap forming portion **18j** of the spacer protrusion **18g** intervenes between the corners **14h** of the heat treatment vessels **14A** and **14D** adjacent in the orthogonal direction F2 and between the corners **14h** of the heat treatment vessels **14B** and **14C** adjacent in the orthogonal direction F2, to form a gap i.e. an orthogonal-direction (lateral) gap S2 between the left wall **14e** of the heat treatment vessel **14A** and the right wall **14d** of the heat treatment vessel **14D** and between the left wall **14e** of the heat treatment vessel **14B** and the right wall **14d** of the heat treatment vessel **14C**.

(31) In the heat treatment furnace **12** configured as above, when any of the four heat treatment vessels **14A** to **14D** coupled together by the coupler **18** try to meander, the corners **14h** of the heat treatment vessels **14A** to **14D** come into engagement with corresponding ones of the pair of first engagement protrusions **18c** and **18d** and the pair of second engagement protrusions **18e** and **18f**, so that the heat treatment vessels **14A** to **14D** are restrained from being displaced mutually. By sandwiching the spacer protrusion **18g** between the four heat treatment vessels **14A** to **14D**, the four heat treatment vessels **14A** to **14D** are restrained from coming closer to one another, so that the four heat treatment vessels **14A** to **14D** coupled together by the coupler **18** are conveyed inside the heat treatment furnace **12** with the conveying-direction gap S1 and the orthogonal-direction gap S2

formed between the four heat treatment vessels **14A** to **14D**. By virtue of formation of the gaps i.e. the conveying-direction gap **S1** and the orthogonal-direction gap **S2** between the four heat treatment vessels **14A** to **14D**, it is possible to appropriately moderate thermal shock generated by temperature difference between the adjacent heat treatment vessels **14A** to **14D** and to appropriately suppress sticking to one another between the adjacent heat treatment vessels **14A** to **14D**, facilitating separation from one another.

(32) As described above, the coupler **18** for the heat treatment vessels **14A** to **14D** of this embodiment includes: the plate body **18b** smaller in size than the heat treatment vessels **14A** to **14D**; the pair of first engagement protrusions **18c** and **18d** protruding from the plate body **18b** to engage with the inner side of the corners **14h** of the heat treatment vessels **14A** and **14B**; the pair of second engagement protrusions **18e** and **18f** protruding from the plate body **18b** to engage with the inner side of the corners **14h** of the heat treatment vessels **14C** and **14D**; and the spacer protrusion **18g** protruding from the plate body **18b** between the pair of first engagement protrusions **18c** and **18d** and between the pair of second engagement protrusions **18e** and **18f** so as to be sandwiched between the heat treatment vessels **14A** and **14B** with which the pair of first engagement protrusions **18c** and **18d** engage respectively and between the heat treatment vessels **14C** and **14D** with which the pair of second engagement protrusions **18e** and **18f** engage respectively. Hence, when the heat treatment vessels **14A** to **14D** try to meander, the corners **14h** of the heat treatment vessels **14A** to **14D** come into engagement with corresponding ones of the pair of first engagement protrusions **18c** and **18d** and the pair of second engagement protrusions **18e** and **18f**, so that the heat treatment vessels **14A** to **14D** are restrained from being displaced mutually, whereby meandering of the heat treatment vessels **14A** to **14D** can be suppressed. By sandwiching the spacer protrusion **18g** between the heat treatment vessels **14A** to **14D**, the heat treatment vessels **14A** to **14D** are restrained from coming closer to one another, so that the heat treatment vessels **14A** to **14D** are conveyed inside the heat treatment furnace **12** with the conveying-direction gap **S1** and the orthogonal-direction gap **S2** formed between the heat treatment vessels **14A** to **14D**, thereby achieving easy separation of the heat treatment vessels **14A** to **14D** after heat treatment, as well as suppression of deterioration in circulation of atmosphere within the heat treatment furnace **12**.

(33) The coupler **18** for the heat treatment vessels **14A** to **14D** of this embodiment includes: a total of four engagement protrusions, i.e., the pair of first engagement protrusions **18c** and **18d** engaging with the corners **14h** of the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction **F1** and the pair of second engagement protrusions **18e** and **18f** engaging with the corners **14h** of the pair of heat treatment vessels **14C** and **14D** adjacent to the pair of heat treatment vessels **14B** and **14A**, respectively, in the orthogonal direction **F2** perpendicular to the conveying direction **F1**; and the spacer protrusion **18g** protruding from the plate body **18b** between the four engagement protrusions so as to be sandwiched between the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction **F1** and between the pair of heat treatment vessels **14C** and **14D** adjacent in the orthogonal direction **F2** perpendicular to the conveying direction **F1**. Thus, mutual displacement is restricted among the total of four heat treatment vessels **14A** to **14D**, i.e., the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction **F1** and the pair of heat treatment vessels **14C** and **14D** adjacent in the orthogonal direction **F2** perpendicular to the conveying direction **F1**, thereby rendering it possible to properly suppress meandering of the heat treatment vessels **14A** to **14D** by the single coupler **18**.

(34) In the coupler **18** for the heat treatment vessels **14A** to **14D** of this embodiment, the spacer protrusion **18g** includes: the conveying-direction gap forming portion **18i** intervening between two ones adjacent in the conveying direction **F1**, of the heat treatment vessels **14A** to **14D**, to form the conveying-direction gap **S1**; and the orthogonal-direction gap forming portion **18j** intervening between two ones adjacent in the orthogonal direction **F2** perpendicular to the conveying direction **F1**, of the heat treatment vessels **14A** to **14D**, to form the orthogonal-direction gap **S2**. Mutual proximity is therefore restricted among the total of four heat treatment vessels **14A** to **14D**, i.e., the

pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction **F1** and the pair of heat treatment vessels **14C** and **14D** adjacent in the orthogonal direction **F2** perpendicular to the conveying direction **F1**, whereupon the heat treatment vessels **14A** to **14D** can be conveyed with the conveying-direction gap **S1** and the orthogonal-direction gap **S2** formed, respectively, between two ones adjacent in the conveying direction **F1** and between two ones adjacent in the orthogonal direction **F2** perpendicular to the conveying direction **F1**, of the heat treatment vessels **14A** to **14D**. (35) In the coupler **18** for the heat treatment vessels **14A** to **14D** of this embodiment, the conveying-direction gap forming portion **18i** has a shape elongated in the orthogonal direction **F2** perpendicular to the conveying direction **F1**, while the orthogonal-direction gap forming portion **18j** has a shape elongated in the conveying direction **F1**, with the spacer protrusion **18g** being a raised portion composed of the conveying-direction gap forming portion **18i** and the orthogonal-direction gap forming portion **18j** that intersect at right angles. Mutual proximity is therefore restricted among the total of four heat treatment vessels **14A** to **14D**, i.e., the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction **F1** and the pair of heat treatment vessels **14C** and **14D** adjacent in the orthogonal direction **F2** perpendicular to the conveying direction **F1**, whereupon the heat treatment vessels **14A** to **14D** can be conveyed with the conveying-direction gap **S1** and the orthogonal-direction gap **S2** formed, respectively, between two ones adjacent in the conveying direction **F1** and between two ones adjacent in the orthogonal direction **F2** perpendicular to the conveying direction **F1**, of the heat treatment vessels **14A** to **14D**.

(36) Another embodiment of the present invention will then be described. In the following description, parts common to the embodiments are designated at the same reference numerals and will not again be described.

Second Embodiment

(37) FIG. 5 is a perspective view explaining a schematic configuration of a heat treatment furnace **100** of another embodiment of the present invention. As shown in FIG. 5, the heat treatment furnace **100** of this embodiment is substantially the same as the heat treatment furnace **12** of the first embodiment except that a pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction **F1** are conveyed in the conveying direction **F1** and that the pair of heat treatment vessels **14A** and **14B** are coupled together by a first coupler (coupler) **102** and a second coupler (coupler) **104** in a pair. The pair of first coupler **102** and second coupler **104** are formed of a sintered body of a heat-resistant inorganic material similar to that of the coupler **18** of the first embodiment.

(38) The first coupler **102** includes integrally, as shown in FIGS. 5 to 7, a plate body **102a**, a pair of engagement protrusions **102b** and **102c**, and a spacer protrusion **102d**. The second coupler **104** includes integrally a plate body **104a**, a pair of engagement protrusions **104b** and **104c**, and a spacer protrusion **104d**. The first coupler **102** and the second coupler **104** are formed in the same shape. FIG. 6 is an enlarged view of one of the pair of first coupler **102** and the second coupler **104** shown in FIG. 5. FIG. 7 is a view of the coupler of FIG. 6, seen from the direction of arrow B.

(39) The plate body **102a** and the plate body **104a** are each a plate member of e.g. a rectangular shape, smaller in size than the bottom walls **14a** of the heat treatment vessels **14A** and **14B**. The pair of first engagement protrusions **102b** and **102c** are two engagement protrusions, as shown in FIGS. 5 to 7, that protrude prismatically from the plate body **102a** at its two corners **102e** so as to engage, respectively, with a corner **14h** of the heat treatment vessel **14B** and with a corner **14h** of the heat treatment vessel **14A**. The engagement protrusion **102b** engages with the corner **14h** between the rear wall **14c** and the right wall **14d** of the heat treatment vessel **14B**, while the engagement protrusion **102c** engages with the corner **14h** between the front wall **14b** and the right wall **14d** of the heat treatment vessel **14A**. The pair of first engagement protrusions **104b** and **104c** are two engagement protrusions, as shown in FIGS. 5 to 7, that protrude prismatically from the plate body **104a** at its two corners **104e** so as to engage, respectively, with a corner **14h** of the heat treatment vessel **14A** and with a corner **14h** of the heat treatment vessel **14B**. The engagement

protrusion **104b** engages with the corner **14h** between the front wall **14b** and the left wall **14e** of the heat treatment vessel **14A**, while the engagement protrusion **104c** engages with the corner **14h** between the rear wall **14c** and the left wall **14e** of the heat treatment vessel **14B**.

(40) The spacer protrusion **102d** is a longitudinal protrusion, as shown in FIGS. 5 to 7, that extends prismatically in the orthogonal direction F2 perpendicular to the conveying direction F1. The spacer protrusion **102d** protrudes from the plate body **102a** between the pair of (two) engagement protrusions **102b** and **102c**, so as to be sandwiched between the corners **14h** of the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1, i.e., so as to be sandwiched between the corner **14h** between the front wall **14b** and the right wall **14d** of the heat treatment vessel **14A** and the corner **14h** between the rear wall **14c** and the right wall **14d** of the heat treatment vessel **14B**.

(41) The spacer protrusion **104d** is a longitudinal protrusion, as shown in FIGS. 5 to 7, that extends prismatically in the orthogonal direction F2 perpendicular to the conveying direction F1. The spacer protrusion **104d** protrudes from the plate body **104a** between the pair of (two) engagement protrusions **104b** and **104c**, so as to be sandwiched between the corners **14h** of the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1, i.e., so as to be sandwiched between: the corner **14h** between the front wall **14b** and the left wall **14e** of the heat treatment vessel **14A**; and the corner **14h** between the rear wall **14c** and the left wall **14e** of the heat treatment vessel **14B**. For this reason, when the pair of first coupler **102** and second coupler **104** are assembled to the corners **14h** of the heat treatment vessels **14A** and **14B** as shown in FIG. 5, the spacer protrusions **102d** and **104d** intervene between corresponding ones of the corners **14h** of the heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1, to form a conveying-direction gap S3 between the front wall **14b** of the heat treatment vessel **14A** and the rear wall **14c** of the heat treatment vessel **14B**.

(42) In the heat treatment furnace **100** configured as above, when the pair of heat treatment vessels **14A** and **14B** coupled together by the pair of first coupler **102** and second coupler **104** try to meander, the corners **14h** of the heat treatment vessels **14A** and **14B** come into engagement with corresponding ones of the pair of engagement protrusions **102b** and **102c** of the first coupler **102** and the pair of engagement protrusions **104b** and **104c** of the second coupler **104**, so that the heat treatment vessels **14A** and **14B** are restrained from being displaced mutually. By sandwiching the spacer protrusions **102d** and **104d** between the pair of heat treatment vessels **14A** and **14B**, the pair of heat treatment vessels **14A** and **14B** are restrained from coming closer to each other, so that the pair of heat treatment vessels **14A** and **14B** coupled together by the pair of first coupler **102** and second coupler **104** are conveyed inside the heat treatment furnace **100** with the conveying-direction gap S3 formed between the pair of heat treatment vessels **14A** and **14B**.

(43) As described above, in the first coupler **102** for the heat treatment vessels **14A** and **14B** of this embodiment, the pair of engagement protrusions **102b** and **102c** are two engagement protrusions that engage with corresponding ones of the corners **14h** of the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1, with the spacer protrusion **102d** projecting from the plate body **102a** between the two engagement protrusions so as to be sandwiched between the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1. This restricts mutual displacement of the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1, whereupon while suppressing meandering of those heat treatment vessels **14** in tandem, the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1 can be conveyed with the conveying-direction gap S3 formed between the pair of heat treatment vessels **14A** and **14B** adjacent in the conveying direction F1.

Third Embodiment

(44) FIG. 8 is a perspective view explaining a configuration of a coupler **110** disposed in a heat treatment furnace of another embodiment of the present invention. The heat treatment furnace of this embodiment is substantially the same as the heat treatment furnace **12** of the first embodiment

except that the coupler **110** differs in shape from the coupler **18** of the first embodiment. The coupler **110** is formed of a sintered body of a heat-resistant inorganic material similar to that of the coupler **18** of the first embodiment.

(45) The coupler **110** includes integrally, as shown in FIGS. **8** and **9**, a plate body **110a**, a pair of first engagement protrusions (engagement protrusions) **110b** and **110c**, a pair of second engagement protrusions (engagement protrusions) **110d** and **110e**, and a spacer protrusion **110f**. The plate body **110a** is a plate member of e.g. a square shape, smaller in size than the bottom walls **14a** of the heat treatment vessels **14A** to **14D**. The pair of first engagement protrusions **110b** and **110c** have the same function as that of the pair of first engagement protrusions **18c** and **18d** of the first embodiment and protrude arcuately in section from the plate body **110a** at its two corners **110g**, to engage respectively with a corresponding one of corners **14h** of the heat treatment vessel **14A** and with a corresponding one of corners **14h** of the heat treatment vessel **14B**. The pair of second engagement protrusions **110d** and **110e** have the same function as that of the pair of second engagement protrusions **18e** and **18f** of the first embodiment and protrude arcuately in section from the plate body **110a** at its two corners **110g**, to engage respectively with a corresponding one of corners **14h** of the heat treatment vessel **14C** and with a corresponding one of corners **14h** of the heat treatment vessel **14D**. The spacer protrusion **110f** has the same function of that of the spacer protrusion **18g** of the first embodiment and protrudes from the plate body **110a** between the pair of first engagement protrusions **110b** and **110c** and between the pair of second engagement protrusions **110d** and **110e** so as to be sandwiched between corresponding ones of the corners **14h** of the heat treatment vessels **14A** and **14B** with which the pair of first engagement protrusions **110b** and **110c** engage and between corresponding ones of the corners **14h** of the heat treatment vessels **14C** and **14D** with which the pair of second engagement protrusions **110d** and **110e** engage. FIG. **9** is a view of the coupler **110** of FIG. **8**, seen from the direction of arrow C.

Fourth Embodiment

(46) FIG. **10** is a perspective view explaining a configuration of a heat treatment vessel **114** disposed in a heat treatment furnace of another embodiment of the present invention. The heat treatment furnace of this embodiment is substantially the same as the heat treatment furnace **12** of the first embodiment except that the heat treatment vessel **114** differs in shape from the heat treatment vessel **14** of the first embodiment. The heat treatment vessel **114** is formed, as shown in FIGS. **1** and **10**, by removing the front wall **14b**, the rear wall **14c**, the right wall **14d**, and the left wall **14e**, other than the four corners **14h**, from the heat treatment vessel **14** of the first embodiment.

Fifth Embodiment

(47) FIG. **11** is a perspective view explaining a schematic configuration of the continuous conveyor type heat treatment furnace **12** to which a coupler **188** of an example of the present invention is properly applied.

(48) The configuration of the coupler **188** will be described with the four heat treatment vessels **14** of FIG. **11** identified as the heat treatment vessels **14A** to **14D**. The heat treatment vessel **14A** is the heat treatment vessel **14** arranged on the leftmost side in the conveying direction F1, of the four heat treatment vessels **14**. The heat treatment vessel **14B** is the heat treatment vessel **14** adjacent to the heat treatment vessel **14A** in the orthogonal direction F2, i.e., the heat treatment vessel **14** arranged on the right side of the heat treatment vessel **14A** in the conveying direction F1. The heat treatment vessel **14C** is the heat treatment vessel **14** arranged on the right side of the heat treatment vessel **14B** in the conveying direction F1. The heat treatment vessel **14D** is the heat treatment vessel **14** arranged on the right side of the heat treatment vessel **14C** in the conveying direction F1, i.e., the heat treatment vessel **14** arranged on the rightmost side in the conveying direction F1, of the four heat treatment vessels **14**. The four heat treatment vessels **14** are disposed, as shown in FIG. **11**, with the coupler **188** coupling the heat treatment vessels **14A** and **14B** together, the coupler **188** coupling the heat treatment vessels **14B** and **14C** together, and the coupler **188**

coupling the heat treatment vessels **14C** and **14D** together, the three couplers **188** each being of the same shape. The configuration of the coupler **188** will hereinafter be described using the coupler **188** that couples together the heat treatment vessel (heat treatment vessel on one hand) **14A** and the heat treatment vessel (heat treatment vessel on the other) **14B** adjacent in the orthogonal direction **F2**.

(49) The coupler **188** includes integrally, as shown in FIGS. **11** to **13**, a plate body **188a**, a pair of engagement protrusions **188b** and **188c**, and a spacer protrusion **188d**. FIG. **12** is an enlarged view of the coupler **188** coupling together the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2** in FIG. **11**, while FIG. **13** is a view of the coupler **188** of FIG. **12**, seen from the direction of arrow **A**. As shown in FIG. **11**, the plate body **188a** is, for example, a square plate member sufficiently smaller in size than the bottom walls **14a** of the heat treatment vessels **14A** and **14B**, the plate body **188a** having sides less than or equal to one-third the length of one side of the heat treatment vessels **14A** and **14B**. As shown in FIGS. **11** to **13**, the engagement protrusion **188b** on one hand of the pair of engagement protrusions **188b** and **188c** protrudes integrally from the plate body **188a** and engages with the right wall **14d** that is one side wall lying toward the heat treatment vessel **14B**, of the front wall **14b**, rear wall **14c**, right wall **14d**, and left wall **14e** of the heat treatment vessel **14A**. As shown in FIGS. **11** to **13**, the engagement protrusion **188c** on the other of the pair of engagement protrusions **188b** and **188c** protrudes integrally from the plate body **188a** and engages with the left wall **14e** that is the other side wall confronting the right wall **14d** of the heat treatment vessel **14A**, of the front wall **14b**, rear wall **14c**, right wall **14d**, and left wall **14e** of the heat treatment vessel **14B**. As shown in FIGS. **11** to **13**, the spacer protrusion **188d** protrudes from the plate body **188a** between the pair of engagement protrusions **188b** and **188c** so as to be sandwiched between the right wall **14d** of the heat treatment vessel **14A** and the left wall **14e** of the heat treatment vessel **14B**.

(50) The engagement protrusion **188b** on one hand of the pair of engagement protrusions **188b** and **188c** is a protrusion, as shown in FIGS. **11** to **13**, that is elongated in the conveying direction **F1** and that protrudes on the plate body **188a** from a side edge **188e** lying toward the heat treatment vessel **14A** so as to face the right wall **14d** of the heat treatment vessel **14A**. The engagement protrusion **188c** on the other of the pair of engagement protrusions **188b** and **188c** is a protrusion, as shown in FIGS. **11** to **13**, that is elongated in the conveying direction **F1** and that protrudes on the plate body **188a** from a side edge **188f** lying toward the heat treatment vessel **14B** so as to face the right wall **14d** of the heat treatment vessel **14B**. The spacer protrusion **188d** is a protrusion, as shown in FIGS. **11** to **13**, that is elongated in the conveying direction **F1** and that protrudes on the plate body **188a** between the pair of engagement protrusions **188b** and **188c** and between the right wall **14d** of the heat treatment vessel **14A** and the left wall **14e** of the heat treatment vessel **14B**.

(51) According to the thus configured heat treatment furnace **12**, if the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2** are each about to progress differently, the right wall **14d** of the heat treatment vessel **14A** or the left wall **14e** of the heat treatment vessel **14B** comes into engagement with a corresponding one of the pair of engagement protrusions **188b** and **188c** so that mutual displacement is restricted between the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2**. For example, if the progress of the heat treatment vessel **14A** is going to be delayed with respect to the heat treatment vessel **14B**, the right wall **14d** of the heat treatment vessel **14A** comes into engagement with the engagement protrusion **188b** of the coupler **188** to generate a frictional force between the right wall **14d** of the heat treatment vessel **14A** and the engagement protrusion **188b**, which frictional force restricts the relative displacement of the heat treatment vessel **14A** in the conveying direction **F1** with respect to the heat treatment vessel **14B**. Since the heat treatment vessel **14** and the coupler **188** for use in this embodiment are made of a porous inorganic material and have rough surfaces, a relatively large frictional force occurs between the heat treatment vessel **14** and the coupler **188**. Since the spacer protrusion **188d** is sandwiched between the right wall **14d** of the heat treatment vessel **14A** and the left wall **14e** of

the heat treatment vessel **14B**, the close proximity is restricted between the heat treatment vessels **14A** and **14B**, whereupon the heat treatment vessels **14A** and **14B** are conveyed inside the heat treatment furnace **12** with an orthogonal-direction gap **S4** formed between the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2**. The orthogonal-direction gap **S4** allows circulation of atmosphere between above and below the heat treatment vessels **14A** and **14B**.

(52) As described above, the coupler **188** for the heat treatment vessels **14A** to **14D** of this embodiment includes: the plate body **188a** smaller than the heat treatment vessels **14A** and **14B**; the pair of engagement protrusions **188b** and **188c** protruding integrally from the plate body **188a** and engaging respectively with the right wall **14d**, lying toward the heat treatment vessel **14B** on the other, of the side walls **14b** to **14e** of the heat treatment vessel **14A** on one hand, of the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2** and with the left wall **14e**, facing the right wall **14d** of the heat treatment vessel **14A**, of the side walls **14b** to **14e** of the heat treatment vessel **14B** on the other; and the spacer protrusion **188d** protruding from the plate body **188a** between the pair of engagement protrusions **188b** and **188c** so as to be sandwiched between the right wall **14d** of the heat treatment vessel **14A** on one hand and the left wall **14e** of the heat treatment vessel **14B** on the other. In consequence, when the heat treatment vessels **14A** and **14B** are each about to progress differently, the right wall **14d** of the heat treatment vessel **14A** or the left wall **14e** of the heat treatment vessel **14B** comes into engagement with a corresponding one of the pair of engagement protrusions **188b** and **188c** so that mutual displacement is restricted between the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2**, thus enabling progression variability to be reduced between the heat treatment vessels **14A** and **14B**. Since the spacer protrusion **188d** is sandwiched between the right wall **14d** of the heat treatment vessel **14A** and the left wall **14e** of the heat treatment vessel **14B**, the intimate proximity is restricted between the heat treatment vessels **14A** and **14B**, whereupon the heat treatment vessels **14A** and **14B** are conveyed inside the heat treatment furnace **12** with the orthogonal-direction gap **S4** formed between the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2**, thus achieving suppression of deterioration in circulation of atmosphere within the heat treatment furnace **12**.

(53) In the coupler **188** for the heat treatment vessels **14A** to **14D** of this embodiment, the pair of engagement protrusions **188b** and **188c** and the spacer protrusion **188d** are each of a shape elongated in the conveying direction **F1** and are parallel to one another. Consequently, the elongated spacer protrusion **188d** can properly restrict the close proximity between the heat treatment vessels **14A** and **14B**.

(54) The coupler **188** for the heat treatment vessels **14A** to **14D** of this embodiment serves to couple together at least three heat treatment vessels **14** adjacent in the orthogonal direction **F2**, i.e., the four heat treatment vessels **14A** to **14D**. This can properly reduce progression variability of the three or more heat treatment vessels **14**, i.e., the four heat treatment vessels **14A** to **14D**, which tend to progress differently from one another.

Sixth Embodiment

(55) FIG. **14** is a perspective view explaining another example of application of the coupler **188** of the present invention. In the heat treatment furnace **12**, as shown in FIG. **14**, a pair of couplers **188** couple together the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2**; another pair of couplers **188** couple together the heat treatment vessels **14B** and **14C** adjacent in the orthogonal direction **F2**; and a further pair of couplers **188** couple together the heat treatment vessels **14C** and **14D** adjacent in the orthogonal direction **F2**.

Seventh Embodiment

(56) FIG. **15** is a perspective view explaining a further example of application of the coupler **188** of the present invention. In the heat treatment furnace **12**, as shown in FIG. **15**, the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2** are conveyed in the conveying

direction F1, coupled together by use of a single coupler **188**.

Eighth Embodiment

(57) FIG. **16** is a perspective view explaining an example of application of a coupler **202** of another embodiment of the present invention. As shown in FIG. **16**, the heat treatment furnace **12** is substantially the same as that of the first embodiment except that the coupler **202** has a different shape. The four heat treatment vessels **14** are disposed, as shown in FIG. **16**, with a pair of couplers **202** coupling the heat treatment vessels **14A** and **14B** together, a pair of couplers **202** coupling the heat treatment vessels **14B** and **14C** together, and a pair of couplers **202** coupling the heat treatment vessels **14C** and **14D** together, the respective pairs of couplers **202** having the same shape. The configuration of the pair of couplers **202** will hereinafter be described using the pair of couplers **202** that couples together the heat treatment vessel (heat treatment vessel on one hand) **14A** and the heat treatment vessel (heat treatment vessel on the other) **14B** adjacent in the orthogonal direction F2. The pair of couplers **202** are formed of a sintered body of a heat-resistant inorganic material similar to that of the coupler **18** of the first embodiment.

(58) The coupler **202** includes integrally, as shown in FIGS. **16** to **18**, a plate body **202a**, a pair of engagement protrusions **202b** and **202c**, a spacer protrusion **202d**, and a misalignment prevention protrusion **202g**. FIG. **17** is an enlarged view of one of the pair of couplers **202** coupling together the heat treatment vessel **14A** and the heat treatment vessel **14B** adjacent in the orthogonal direction F2 in FIG. **16**, while FIG. **18** is a view of the coupler **202** of FIG. **17**, seen from the direction of arrow B. As shown in FIG. **16**, the plate body **202a** is, for example, a rectangular plate member extending in the orthogonal direction F2, sufficiently smaller in size than the bottom walls **14a** of the heat treatment vessels **14A** and **14B**. As shown in FIGS. **17** and **18**, the engagement protrusion **202b** on one hand of the pair of engagement protrusions **202b** and **202c** protrudes integrally prismatically from the plate body **202a** at its corner **202f**. As shown in FIG. **16**, the engagement protrusion **202b** engages with an end, toward the front wall **14b**, of the right wall **14d** that is one side wall, lying toward the heat treatment vessel **14B**, of the front wall **14b**, rear wall **14c**, right wall **14d**, and left wall **14e** of the heat treatment vessel **14A**, i.e., with a corner **14f** formed between the right wall **14d** and the front wall **14b** of the heat treatment vessel **14A**. As shown in FIGS. **17** and **18**, the engagement protrusion **202c** on the other of the pair of engagement protrusions **202b** and **202c** protrudes integrally prismatically from the plate body **202a** at its corner **202f**. As shown in FIG. **16**, the engagement protrusion **202c** engages with an end, toward the front wall **14b**, of the left wall **14e** that is the other side wall, facing the right wall **14d** of the heat treatment vessel **14A**, of the front wall **14b**, rear wall **14c**, right wall **14d**, and left wall **14e** of the heat treatment vessel **14B**, i.e., with a corner **14f** formed between the left wall **14e** and the front wall **14b** of the heat treatment vessel **14B**. The spacer protrusion **202d** protrudes from the plate body **202a** between the pair of engagement protrusions **202b** and **202c** so as to be sandwiched between the right wall **14d** of the heat treatment vessel **14A** and the left wall **14e** of the heat treatment vessel **14B**. The spacer protrusion **202d** is elongated in the conveying direction F1. The misalignment prevention protrusion **202g** protrudes from the plate body **202a** at its side edge not having the pair of engagement protrusions **202b** and **202c**, of both side edges of the plate body **202a**, in the same direction as the direction where the pair of engagement protrusions **202b** and **202c** protrude. The misalignment prevention protrusion **202g** is joined at its central portion in the longitudinal direction to the spacer protrusion **202d**. Between the misalignment prevention protrusion **202g** and the pair of engagement protrusions **202b** and **202c** there is an interval greater than the thickness of the front wall **14b** and the rear wall **14c**.

(59) According to the thus configured coupler **202**, if the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction F2 are each about to progress differently, the corner **14f** of the heat treatment vessel **14A** or the corner **14f** of the heat treatment vessel **14B** abuts against a corresponding one of the pair of engagement protrusions **202b** and **202c** of the coupler **202**, so that mutual displacement is restricted between the heat treatment vessels **14A** and **14B** adjacent in the

orthogonal direction F2. For example, if the progress of the heat treatment vessel **14A** is going to be delayed with respect to the heat treatment vessel **14B**, the corner **14f** formed between the right wall **14d** and the front wall **14b** of the heat treatment vessel **14A** abuts against the engagement protrusion **202b** of the coupler **202**, thereby restricting the displacement of the heat treatment vessel **14A** relative to the heat treatment vessel **14B**. Since the spacer protrusion **202d** is sandwiched between the right wall **14d** of the heat treatment vessel **14A** and the left wall **14e** of the heat treatment vessel **14B**, the close proximity is restricted between the heat treatment vessels **14A** and **14B** so that the heat treatment vessels **14A** and **14B** are conveyed inside the heat treatment furnace **12** with the orthogonal-direction gap **S4** formed between the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction F2. The misalignment prevention protrusion **202g** prevents misalignment of the couplers **202** by sandwiching the front wall **14b** and the rear wall **14c** between the misalignment prevention protrusion **202g** and a corresponding one of the pair of engagement protrusions **202b** and **202c**. Since the misalignment prevention protrusion **202g** lies outside of the front wall **14b** and the rear wall **14c**, a gap is formed between one pair of heat treatment vessels **14A** and **14B** and the other pair of heat treatment vessels **14A** and **14B**, adjacent in the conveying direction F1, to thereby secure vertical circulation of in-furnace atmosphere.

(60) As described above, according to the coupler **202** for the heat treatment vessels **14A** to **14D** of this embodiment, the pair of engagement protrusions **202b** and **202c** engage respectively with an end of the right wall **14d** of the heat treatment vessel **14A** and with an end of the left wall **14e** of the heat treatment vessel **14B**. For this reason, if the heat treatment vessels **14A** and **14B** are each about to progress differently, the corner **14f** of the heat treatment vessel **14A** or **14B** comes into abutment against a corresponding one of the pair of engagement protrusions **202b** and **202c**, so that mutual displacement can properly be restricted between the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction F2.

(61) The coupler **202** for the heat treatment vessels **14A** to **14D** of this embodiment includes the misalignment prevention protrusion **202g** that protrudes from the plate body **202a** at its side edge not having the pair of engagement protrusions **202b** and **202c**, of both side edges of the plate body **202a**, in the same direction as the direction where the pair of engagement protrusions **202b** and **202c** protrude, the misalignment prevention protrusion **202g** lying with an interval greater than the thickness of the front wall **14b** and the rear wall **14c** of the heat treatment vessels **14A** and **14B** between the misalignment prevention protrusion **202g** and the pair of engagement protrusions **202b** and **202c**. Hence, the misalignment prevention protrusion **202g** can prevent misalignment of the couplers **202** by sandwiching the front wall **14b** and the rear wall **14c** of each of the heat treatment vessels **14A** and **14B** between the misalignment prevention protrusion **202g** and the pair of engagement protrusions **202b** and **202c**.

Ninth Embodiment

(62) FIG. **19** is a perspective view explaining an example of application of a coupler **302** of another embodiment of the present invention. The four heat treatment vessels **14** are disposed, as shown in FIG. **19**, with a coupler **302** coupling the heat treatment vessels **14A** and **14B** together, a coupler **302** coupling the heat treatment vessels **14B** and **14C** together, and a coupler **302** coupling the heat treatment vessels **14C** and **14D** together, those three couplers **302** having the same shape. The configuration of the coupler **302** will hereinafter be described using the coupler **302** that couples together the heat treatment vessel (heat treatment vessel on one hand) **14A** and the heat treatment vessel (heat treatment vessel on the other) **14B** adjacent in the orthogonal direction F2. The coupler **302** is formed of a sintered body of a heat-resistant inorganic material similar to that of the coupler **18** of the first embodiment.

(63) The coupler **302** includes integrally, as shown in FIGS. **19** to **21**, a plate body **302a**, a pair of engagement protrusions **302b** and **302c**, and a spacer protrusion **302d**. FIG. **20** is an enlarged view of the coupler **302** coupling together the heat treatment vessel **14A** and the heat treatment vessel **14B** adjacent in the orthogonal direction F2 in FIG. **19**, while FIG. **21** is a view of the coupler **302**

of FIG. 19, seen from the direction of arrow C. As shown in FIG. 19, the plate body 302a is, for example, a rectangular plate member extending in the conveying direction F1, slightly smaller in size than the bottom walls 14a of the heat treatment vessels 14A and 14B, i.e., a long plate having a length along the right wall 14d of the heat treatment vessel 14A and the left wall 14e of the heat treatment vessel 14B between both ends of each of the right wall 14d and the left wall 14e. The engagement protrusion 302b on one hand of the pair of engagement protrusions 302b and 302c protrudes integrally from the plate body 302a as shown in FIGS. 20 and 21 and engages with the right wall 14d that is one side wall, lying toward the heat treatment vessel 14B, of the front wall 14b, rear wall 14c, right wall 14d, and left wall 14e of the heat treatment vessel 14A as shown in FIG. 19. The engagement protrusion 302c on the other of the pair of engagement protrusions 302b and 302c protrudes integrally from the plate body 302a as shown in FIGS. 20 and 21 and engages with the left wall 14e that is the other side wall, facing the right wall 14d of the heat treatment vessel 14A, of the front wall 14b, rear wall 14c, right wall 14d, and left wall 14e of the heat treatment vessel 14B as shown in FIG. 19. The spacer protrusion 302d protrudes from the plate body 302a between the pair of engagement protrusions 302b and 302c so as to be sandwiched between the right wall 14d of the heat treatment vessel 14A and the left wall 14e of the heat treatment vessel 14B as shown in FIGS. 19 to 21.

(64) The engagement protrusion 302b on one hand of the pair of engagement protrusions 302b and 302c is a protrusion elongated in the conveying direction F1, that projects on the plate body 302a from the side edge 302e, lying toward the heat treatment vessel 14A, of the plate body 302a so as to face the right wall 14d of the heat treatment vessel 14A as shown in FIGS. 19 to 21. The engagement protrusion 302c on the other of the pair of engagement protrusions 302b and 302c is a protrusion elongated in the conveying direction F1, that projects on the plate body 302a from the side edge 302f, lying toward the heat treatment vessel 14B, of the plate body 302a so as to face the left wall 14e of the heat treatment vessel 14B as shown in FIGS. 19 to 21. The spacer protrusion 302d is a protrusion elongated in the conveying direction F1, that projects from the plate body 302a between the pair of engagement protrusions 302b and 302c and between the right wall 14d of the heat treatment vessel 14A and the left wall 14e of the heat treatment vessel 14B as shown in FIGS. 19 to 21.

(65) For example, if the heat treatment vessels 14A and 14B adjacent in the orthogonal direction F2 are each about to progress differently, the right wall 14d of the heat treatment vessel 14A or the left wall 14e of the heat treatment vessel 14B comes into engagement with a corresponding one of the pair of engagement protrusions 302b and 302c, so that mutual displacement is restricted between the heat treatment vessels 14A and 14B adjacent in the orthogonal direction F2. For example, if the progress of the heat treatment vessel 14A is going to be delayed with respect to the heat treatment vessel 14B, the right wall 14d of the heat treatment vessel 14A comes into engagement with the engagement protrusion 302b of the coupler 302 to generate a frictional force between the right wall 14d of the heat treatment vessel 14A and the engagement protrusion 302b, which frictional force restricts the displacement of the heat treatment vessel 14A relative to the heat treatment vessel 14B. If the heat treatment vessels 14A and 14B adjacent in the orthogonal direction F2 are displaced mutually, the corner 14f of the heat treatment vessel 14A or 14B comes into abutment against a corresponding one of the pair of engagement protrusions 302b and 302c. Since the spacer protrusion 302d is sandwiched between the right wall 14d of the heat treatment vessel 14A and the left wall 14e of the heat treatment vessel 14B, the close proximity is restricted between the heat treatment vessels 14A and 14B, whereupon the heat treatment vessels 14A and 14B are conveyed inside the heat treatment furnace 12 with the orthogonal-direction gap S4 formed between the heat treatment vessels 14A and 14B adjacent in the orthogonal direction F2.

(66) As described above, according to the coupler 302 for the heat treatment vessels 14A to 14D of this embodiment, in the state where the coupler 302 couples the adjacent vessels together, the plate body 302a is a long plate extending along the right wall 14d of the heat treatment vessel 14A and

the left wall **14e** of the heat treatment vessel **14B**, with the engagement protrusion **302b** on one hand of the pair of engagement protrusions **302b** and **302c** being an elongated protrusion that projects, on the plate body **302a** that is the long plate, from the side edge **302e**, lying toward the heat treatment vessel **14A**, of the plate body **302** so as to face the right wall **14d** of the heat treatment vessel **14A**, and with the engagement protrusion **302c** on the other of the pair of engagement protrusions **302b** and **302c** being an elongated protrusion that projects, on the plate body **302a** that is the long plate, from the side edge **302f**, lying toward the heat treatment vessel **14B**, of the plate body **302** so as to face the left wall **14e** of the heat treatment vessel **14B**. For this reason, if the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2** are displaced relative to each other, the corner **14f** of the heat treatment vessel **14A** or **14B** comes into abutment against a corresponding one of the pair of engagement protrusions **302b** and **302c** so that mutual displacement can properly be restricted between the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2**.

(67) Although the embodiments of the present invention have hereinbefore been described in detail based on the drawings, the present invention is applicable also to other modes.

(68) Although, for example, in the coupler **18** of the first embodiment described above, the pair of first engagement protrusions **18c** and **18d** and the pair of second engagement protrusions **18e** and **18f** each protrude prismatically, for example, the pair of first engagement protrusions **18c** and **18d** and the pair of second engagement protrusions **18e** and **18f** may protrude cylindrically. That is, the pair of first engagement protrusions **18c** and **18d** and the pair of second engagement protrusions **18e** and **18f** may be of any shape as long as they can engage with corresponding ones of the corners **14h** of the pair of heat treatment vessels **14A** and **14B** and the corners **14h** of the pair of heat treatment vessels **14C** and **14D**. Although the conveying-direction gap forming portion **18i** and the orthogonal-direction gap forming portion **18j** of the spacer protrusion **18g** are each a protrusion that extends prismatically, for example, the conveying-direction gap forming portion **18i** and the orthogonal-direction gap forming portion **18j** may each be a protrusion that extends cylindrically. That is, the conveying-direction gap forming portion **18i** and the orthogonal-direction gap forming portion **18j** may be of any shape as long as they can form the conveying-direction gap **S1** and orthogonal-direction gap **S2** between adjacent ones of the heat treatment vessels **14A** to **14D**.

(69) Although the spacer protrusion **18g** of the coupler **18** of the first embodiment described above has both the conveying-direction gap forming portion **18i** and the orthogonal-direction gap forming portion **18j**, for example, the spacer protrusion **18g** may have only one of the conveying-direction gap forming portion **18i** and the orthogonal-direction gap forming portion **18j**. That is, circulation of atmosphere within the heat treatment furnace **12** can be restrained from deteriorating, due to the formation of either gap of the conveying-direction gap **S1** and the orthogonal-direction gap **S2** between adjacent ones of the heat treatment vessels **14A** to **14D**.

(70) Although in the second embodiment described above, the pair of heat treatment vessels **14A** and **14B** are coupled together by the pair of first coupler **102** and second coupler **104**, for example, the pair of heat treatment vessels **14A** and **14B** may be coupled together by only either coupler of the pair of first coupler **102** and the second coupler **104**.

(71) Although in the coupler **188** of the fifth to seventh embodiments described above, the pair of engagement protrusions **188b** and **188c** are each of a shape elongated in the conveying direction **F1**, for example, they may extend in a prismatic shape or a cylindrical shape. That is, the pair of engagement protrusions **188b** and **188c** may extend in any shape as long as the pair of engagement protrusions **188b** and **188c** are allowed to engage with the right wall **14d** of the heat treatment vessel **14A** and the left wall **14e** of the heat treatment vessel **14B**. Although the spacer protrusion **188d** has a shape elongated in the conveying direction **F1**, similarly, it may extend in a prismatic shape or a cylindrical shape. That is, the spacer protrusion **188d** may extend in any shape as long as the spacer protrusion **188d** can be sandwiched between the right wall **14d** of the heat treatment vessel **14A** and the left wall **14e** of the heat treatment vessel **14B**.

(72) Although in the coupler **202** of the eighth embodiment described above, the pair of engagement protrusions **202b** and **202c** each protrude in a prismatic shape, for example, the pair of engagement protrusions **202b** and **202c** may each extend in a cylindrical shape. That is, the pair of engagement protrusions **202b** and **202c** may extrude in any shape as long as the pair of engagement protrusions **202b** and **202c** can engage with the end of the right wall **14d** of the heat treatment vessel **14A** and the end of the left wall **14e** of the heat treatment vessel **14B**, i.e., as long as the pair of engagement protrusions **202b** and **202c** can engage with the corners **14f** of the heat treatment vessels **14A** and **14B**. The misalignment prevention protrusion **202g** and the spacer protrusion **302d** need not necessarily be joined together.

(73) Although in the eighth embodiment described above, the heat treatment vessels **14A** and **14B** are coupled together by the pair of couplers **202**, for example, the heat treatment vessels **14A** and **14B** may be coupled together by only one coupler of the pair of couplers **202**.

(74) Although the coupler **188** shown in FIGS. **11** to **15** of the first to third embodiments described above couples together the heat treatment vessels **14A** and **14B** adjacent in the orthogonal direction **F2**, it may be used to couple together the heat treatment vessels **14** adjacent in the conveying direction **F1**.

(75) It should be appreciated that the above are merely exemplary embodiments and that the present invention can be carried out in modes variously altered or modified based on the knowledge of those skilled in the art.

EXPLANATIONS OF LETTERS OR NUMERALS

(76) **10**: material to be treated **12**, **100**: heat treatment furnace **14**, **14A**, **14B**, **14C**, **14D**, **114**: heat treatment vessel **14b**: front wall (side wall) **14c**: rear wall (side wall) **14d**: right wall (side wall) **14e**: left wall (side wall) **14h**: corner **16**: conveyor roller **18**, **110**: coupler **18b**, **110a**: plate body **18c**, **18d**, **110b**, **110c**: first engagement protrusion (engagement protrusion) **18e**, **18f**, **110d**, **110e**: second engagement protrusion (engagement protrusion) **18g**, **110f**: spacer protrusion **18i**: conveying-direction gap forming portion **18j**: orthogonal-direction gap forming portion **102**: first coupler (coupler) **102a**: plate body **102b**, **102c**: engagement protrusion **102d**: spacer protrusion **104**: second coupler (coupler) **104a**: plate body **104b**, **104c**: engagement protrusion **104d**: spacer protrusion **188**, **202**, **302**: coupler **188a**, **202a**, **302a**: plate body **188b**, **188c**, **202b**, **202c**, **302b**, **302c**: engagement protrusion **188d**, **202d**, **302d**: spacer protrusion **202g**: misalignment prevention protrusion **302e**, **302f**: side edge **F1**: conveying direction **F2**: orthogonal direction (direction) **S1**: conveying-direction gap (gap) **S2**: orthogonal-direction gap (gap)

Claims

1. A heat treatment vessel coupler for coupling together heat treatment vessels adjacent in a horizontal direction, of a plurality of rectangular box-shaped heat treatment vessels each loaded with a material to be treated, conveyed in a conveying direction on a plurality of rotationally driven conveyor rollers arranged in parallel inside a heat treatment furnace, the coupler comprising: a plate body smaller than each of the heat treatment vessels; at least a pair of engagement protrusions each projecting from the plate body and engaging with a corresponding one of side walls of the heat treatment vessels; and a spacer protrusion projecting from the plate body between the at least a pair of engagement protrusions so as to be sandwiched between the heat treatment vessels with which the at least a pair of engagement protrusions engage respectively.

2. The heat treatment vessel coupler of claim 1, wherein the coupler couples mutually the heat treatment vessels adjacent in the conveying direction, of the plurality of rectangular box-shaped heat treatment vessels or couples mutually the heat treatment vessels adjacent in the conveying direction and mutually the heat treatment vessels adjacent in a direction orthogonal to the conveying direction, wherein at least a pair of engagement protrusions each engage with a corresponding one of corners defined between side walls of the heat treatment vessels, and wherein

- the spacer protrusion projects from the plate body between the at least a pair of engagement protrusions so as to be sandwiched between the corners defined between side walls of the heat treatment vessels with which the at least a pair of engagement protrusions engage respectively.
3. The heat treatment vessel coupler of claim 2, wherein the at least a pair of engagement protrusions include a total of four engagement protrusions, i.e., a pair of engagement protrusions each engaging with a corresponding one of corners defined between side walls of a pair of heat treatment vessels adjacent in the conveying direction and a pair of engagement protrusions each engaging with a corresponding one of corners defined between side walls of a pair of heat treatment vessels adjacent in a direction orthogonal to the conveying direction, and wherein the spacer protrusion projects from the plate body between the four engagement protrusions so as to be sandwiched between the pair of heat treatment vessels adjacent in the conveying direction and between the pair of heat treatment vessels adjacent in the direction orthogonal to the conveying direction.
4. The heat treatment vessel coupler of claim 3, wherein the spacer protrusion comprises: a conveying-direction gap forming portion intervening between heat treatment vessels adjacent in the conveying direction to form a gap therebetween; and an orthogonal-direction gap forming portion intervening between heat treatment vessels adjacent in a direction orthogonal to the conveying direction to form a gap therebetween.
5. The heat treatment vessel coupler of claim 4, wherein the conveying-direction gap forming portion is elongated in a direction orthogonal to the conveying direction, wherein the orthogonal-direction gap forming portion is elongated in the conveying direction, and wherein the spacer protrusion is a protrusion including the conveying-direction gap forming portion and the orthogonal-direction gap forming portion that intersect each other at right angles.
6. The heat treatment vessel coupler of claim 3, comprising: a misalignment prevention protrusion projecting on the plate body from a side edge not having the pair of engagement protrusions, of both side edges of the plate body, to a direction identical to a direction of projection of the pair of engagement protrusions, the misalignment prevention protrusion being positioned, with intervals greater than a thickness of the side walls of the heat treatment vessel, between the pair of engagement protrusions.
7. The heat treatment vessel coupler of claim 2, wherein the at least a pair of engagement protrusions include two engagement protrusions each engaging with a corresponding one of corners defined between side walls of a pair of heat treatment vessels adjacent in the conveying direction, wherein the spacer protrusion projects from the plate body between the two engagement protrusions so as to be sandwiched between the pair of heat treatment vessels adjacent in the conveying direction.
8. The heat treatment vessel coupler of claim 1, wherein the coupler couples mutually heat treatment vessels adjacent in a direction orthogonal to the conveying direction, of the rectangular box-shaped heat treatment vessels, wherein the pair of engagement protrusions project integrally from the plate body and engage respectively with one side wall, lying toward a heat treatment vessel on the other hand, of side walls of a heat treatment vessel on one hand, of the heat treatment vessels adjacent in the orthogonal direction and with the other side wall, facing the one side wall, of side walls of a heat treatment vessel on the other hand, of the heat treatment vessels adjacent in the orthogonal direction, and wherein the spacer protrusion project from the plate body between the pair of engagement protrusions so as to be sandwiched between the one side wall and the other side wall.
9. The heat treatment vessel coupler of claim 8, wherein the pair of engagement protrusions and the spacer protrusion are elongated in the conveying direction and are parallel to one another.
10. The heat treatment vessel coupler of claim 8, wherein the pair of engagement protrusions engage with an end of the one side wall and an end of the other side wall, respectively.
11. The heat treatment vessel coupler of claim 8, wherein in a state where the coupler couples

adjacent vessels together, the plate body is an elongated plate extending along the one side wall and the other side wall, one of the pair of engagement protrusions being a longitudinal protrusion projecting from a side edge, lying toward the heat treatment vessel on one hand, of the elongated plate, so as to face the one side wall, the other of the pair of engagement protrusions being a longitudinal protrusion projecting from a side edge, lying toward the heat treatment vessel on the other hand, of the elongated plate, so as to face the other side wall.

12. A heat treatment vessel coupling method of coupling mutually, by use of a coupler, heat treatment vessels adjacent in a conveying direction or heat treatment vessels adjacent in an orthogonal direction orthogonal to the conveying direction, of rectangular box-shaped heat treatment vessels each loaded with a material to be treated, conveyed on a plurality of rotationally driven conveyor rollers arranged in parallel inside a heat treatment furnace, the coupler comprising: a plate body smaller than each of the heat treatment vessels; a pair of engagement protrusions projecting integrally from the plate body and engaging respectively with one side wall, lying toward a heat treatment vessel on the other hand, of side walls of a heat treatment vessel on one hand, of heat treatment vessels adjacent in the conveying direction or the orthogonal direction and with the other side wall, facing the one side wall, of side walls of the heat treatment vessel on the other hand, of the heat treatment vessels adjacent in the conveying direction or the orthogonal direction, and a spacer protrusion projecting from the plate body between the pair of engagement protrusions so as to be sandwiched between the one side wall and the other side wall.

13. A heat treatment method using a heat treatment vessel coupler, the coupler coupling mutually, heat treatment vessels adjacent in a conveying direction of the heat treatment vessels or heat treatment vessels adjacent in an orthogonal direction orthogonal to the conveying direction, of rectangular box-shaped heat treatment vessels each loaded with a material to be treated, conveyed on a plurality of rotationally driven conveyor rollers arranged in parallel inside a heat treatment furnace, the coupler comprising: a plate body smaller than each of the heat treatment vessels; a pair of engagement protrusions projecting integrally from the plate body and engaging respectively with one side wall, lying toward a heat treatment vessel on the other hand, of side walls of a heat treatment vessel on one hand, of heat treatment vessels adjacent in the conveying direction or the orthogonal direction and with the other side wall, facing the one side wall, of side walls of the heat treatment vessel on the other hand, of the heat treatment vessels adjacent in the conveying direction or the orthogonal direction, and a spacer protrusion projecting from the plate body between the pair of engagement protrusions so as to be sandwiched between the one side wall and the other side wall.
